

FORMULAS FOR REFERENCE

SPHERE	Surface area	$= 4\pi r^2$
	Volume	$= \frac{4}{3}\pi r^3$
CYLINDER	Area of curved surface	$= 2\pi rh$
	Volume	$= \pi r^2 h$
CONE	Area of curved surface	$= \pi rl$
	Volume	$= \frac{1}{3}\pi r^2 h$
PRISM	Volume	$= \text{base area} \times \text{height}$
PYRAMID	Volume	$= \frac{1}{3} \times \text{base area} \times \text{height}$

There are 36 questions in Section A and 18 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.

Section A

Label 1 If $a = 2 - \frac{1}{1+b}$, then $b =$

A. $\frac{1-a}{a-2}$

B. $\frac{a-1}{a-2}$

C. $\frac{a+1}{a-2}$

D. $\frac{-a-3}{a-2}$

E. $\frac{1-a}{a}$

Label 2 $(2x^2 - 3x + 1)(2 - 3x) =$

A. $6x^3 - 5x^2 - 3x + 2$

B. $6x^3 - 13x^2 - 9x - 2$

C. $-6x^3 + 13x^2 - 9x + 2$

D. $-6x^3 - 5x^2 - 3x + 2$

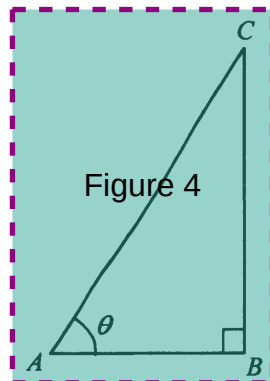
E. $-6x^3 - 5x^2 - 9x + 2$

Label 3 Let $f(x) = (2x-1)(x+1) + 2x+1$. Find the remainder when $f(x)$ is divided by $2x+1$.

- A. $\frac{1}{2}$
- B. $\frac{1}{2}$
- C. 0
- D. 1
- E. 2

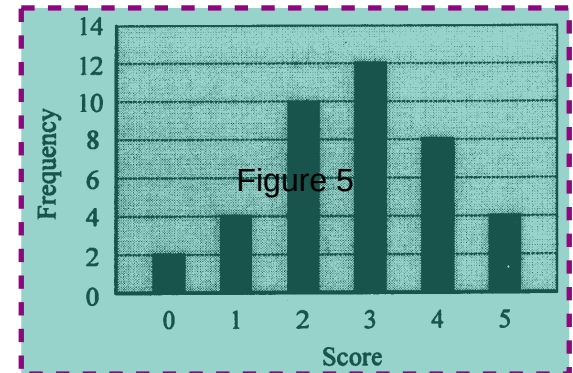
Label 4 The figure shows a right-angled triangle where $AB:BC = 3:4$. Find $\sin \theta$.

- A. $\frac{5}{3}$
- B. $\frac{3}{4}$
- C. $\frac{5}{4}$
- D. $\frac{3}{5}$
- E. $\frac{4}{5}$



Label 5 The bar chart below shows the distribution of scores in a test. Find the percentage of scores which are less than 3.

- A. 35%
- B. 40%
- C. 50%
- D. 65%
- E. 70%

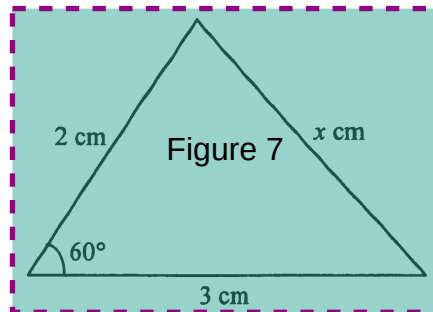


Label 6 If the straight lines $2x-3y+1=0$ and $5x+ky-1=0$ are perpendicular to each other, find k .

- A. $\frac{15}{2}$
- B. $\frac{10}{3}$
- C. $\frac{3}{10}$
- D. $\frac{10}{3}$
- E. $\frac{15}{2}$

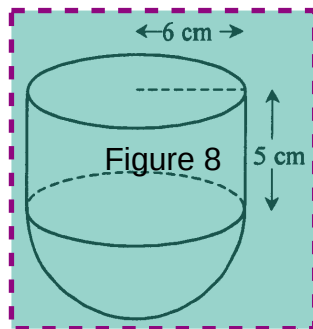
Label 7 In the figure, find x correct to 3 significant figures.

- A. 2.65
- B. 2.79
- C. 3.16
- D. 4.00
- E. 4.36



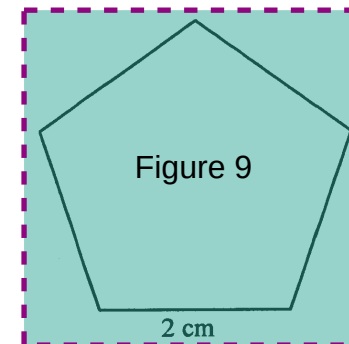
Label 8 In the figure, the solid consists of a cylinder and a hemisphere with a common base of radius 6 cm. Find the total surface area of the solid.

- A. $132\pi \text{ cm}^2$
- B. $168\pi \text{ cm}^2$
- C. $204\pi \text{ cm}^2$
- D. $240\pi \text{ cm}^2$
- E. $324\pi \text{ cm}^2$



Label 9 The figure shows a regular pentagon. Find its area correct to the nearest 0.01 cm^2 .

- A. 3.63 cm^2
- B. 5.88 cm^2
- C. 6.18 cm^2
- D. 6.88 cm^2
- E. 8.51 cm^2



Label 10 $\frac{a^{n-2} + a^{n-1}}{a^{n-2}} =$

- A. a^{n-1}
- B. $a^{n-2}(1+a)$
- C. $1+a^{n-1}$
- D. $1+\frac{1}{a}$
- E. $1+a$

Label 1 Which of the following is an identity / are identities?

- I. $x^2 + 2x + 1 = 0$
- II. $x^2 + 2x + 1 = (x + 1)^2$
- III. $x^2 + 1 > 0$

- A. I only
- B. II only
- C. III only
- D. I and III only
- E. II and III only

Label 12 If $\begin{cases} y = x^2 - 4x - 44 \\ y = -2x + 4 \end{cases}$ then $y =$

- A. -32 or 52
- B. -12 or 16
- C. -12 or 96
- D. -8 or 20
- E. 12 or 24

Label 13 A piece of wire of length 36 cm is cut into two parts. One part, x cm long, is bent into a square and the other part is bent into a circle. If the length of a side of the square is equal to the radius of the circle, which of the following equations can be used to find x ?

- A. $x = \frac{36 - 4x}{2\pi}$
- B. $x = \frac{36 - x}{2\pi}$
- C. $\frac{x}{4} = \frac{36 - 4x}{2\pi}$
- D. $\frac{x}{4} = \frac{36 - x}{\pi}$
- E. $\frac{x}{4} = \frac{36 - x}{2\pi}$

Label 14 The sum of the first n terms of an arithmetic sequence is n^2 . Find the 10th term of the sequence.

- A. 19
- B. 21
- C. 28
- D. 31
- E. 100

Label 15 The n th term of a geometric sequence is $-\frac{1}{2^n}$. Find the first term and the common ratio.

- | | first term | common ratio |
|---|----------------|----------------|
| A | 1 | $\frac{1}{2}$ |
| B | $-\frac{1}{2}$ | $-\frac{1}{2}$ |
| C | $-\frac{1}{2}$ | $\frac{1}{2}$ |
| D | $-\frac{1}{2}$ | 1 |
| E | 1 | $-\frac{1}{2}$ |

Label 16 A bank offers loans at an interest rate of 18% per annum, compounded monthly. A man took a loan of \$20 000 and repays the bank in monthly instalments of \$4 000. Find the outstanding balance after his first instalment.

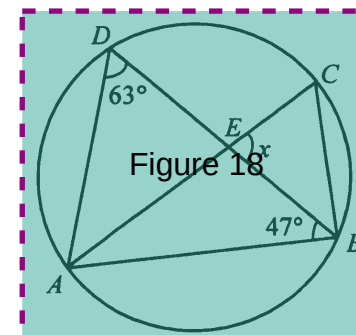
- A \$16 000
 B \$16 240
 C \$16 300
 D \$18 880
 E \$19 600

Label 17 If $0^\circ < x < y < 90^\circ$, which of the following must be true?

- I. $\sin x < \sin y$
 II. $\cos x < \cos y$
 III. $\sin x < \cos y$
- A I only
 B II only
 C I and II only
 D I and III only
 E II and III only

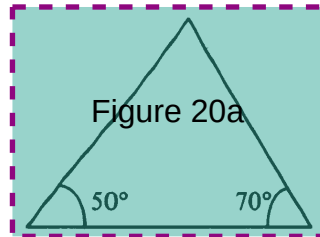
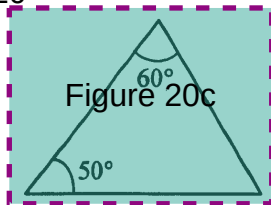
Label 18 In the figure, AEC is a diameter and DEB is a straight line. Find x .

- A 54°
 B 70°
 C 74°
 D 92°
 E 94°

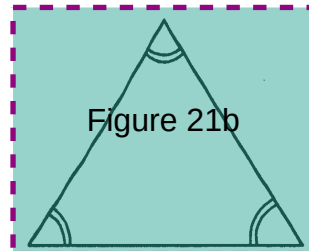
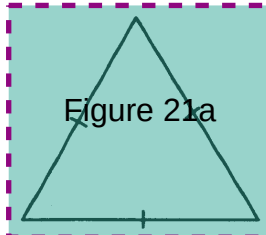


Label 19 Which of the following pairs of triangles is/are similar?

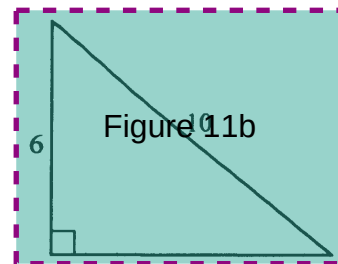
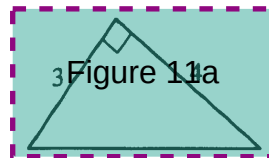
Label 20



Label 21



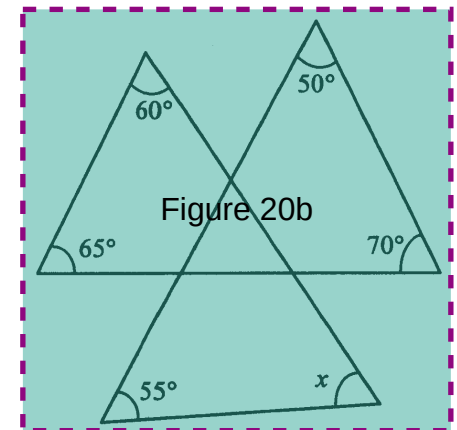
Label 11



- A. II only
- B. III only
- C. I and II only
- D. I and III only
- E. I, II and III

Label 20 In the figure, $x =$

- A. 50°
- B. 55°
- C. 60°
- D. 65°
- E. 70°



Label 21 If the mean of the ten numbers 8, 6, 6, 6, 7, 4, 10, 9, 9, x is 7, find the median of the ten numbers.

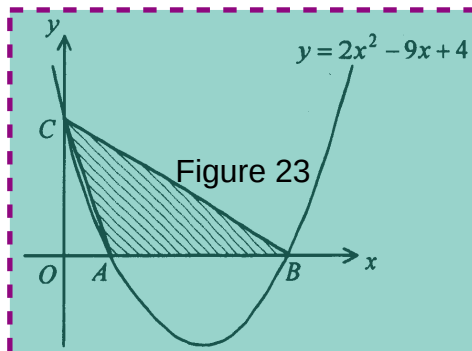
- A. 3.5
- B. 6
- C. 6.5
- D. 7
- E. 7.5

Label 22 Which of the following is a factor of $2(a-b)^2 - a^2 + b^2$?

- A. $a-3b$
- B. $a-2b$
- C. $a+b$
- D. $a+3b$
- E. $3a-b$

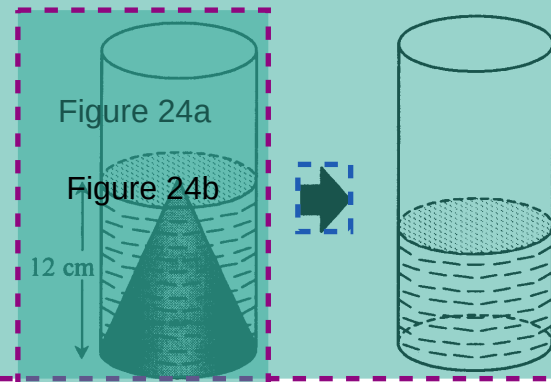
Label 23 In the figure, the graph of $y = 2x^2 - 9x + 4$ cuts the x -axis at A and B , and the y -axis at C . Find the area of $\triangle ABC$.

- A. 4
- B. 6
- C. 7
- D. 8
- E. 14



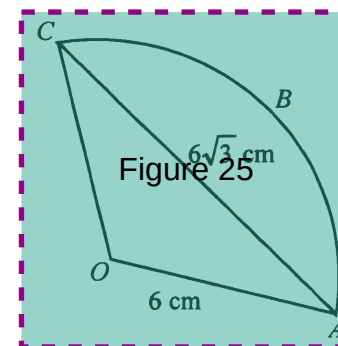
Label 24 In the figure, a solid right circular cone of height 12 cm is put into a cylinder which has the same internal radius as the base radius of the cone. Water is then poured into the cylinder until the water level just reaches the tip of the cone. If the cone is removed, what is the height of the water in the cylinder?

- A. 3 cm
- B. 4 cm
- C. 6 cm
- D. 8 cm
- E. 9 cm



Label 25 In the figure, $OABC$ is a sector. Find the length of the arc ABC .

- A. $\frac{2\pi}{3}$ cm
- B. 4π cm
- C. 3π cm
- D. 6π cm
- E. 12π cm



Label 26 In the figure, A , B and C are the centres of three equal circles, each of radius 1 cm. Find the area of the shaded region.

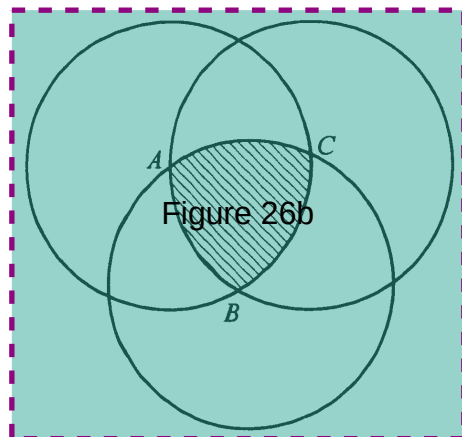
A. $\left(\frac{\sqrt{3}}{2} - \frac{\pi}{4}\right) \text{ cm}^2$

B. $\left(\frac{\sqrt{3}}{2} - \frac{\pi}{4}\right) \text{ cm}^2$

C. $\left(\frac{\sqrt{3}}{2} - \frac{\pi}{4}\right) \text{ cm}^2$

D. $\frac{\pi}{2} \text{ cm}^2$

E. $\left(\frac{\sqrt{3}}{2} - \frac{\pi}{4}\right) \text{ cm}^2$



Label 27 40% of the students in a class failed in a test. They had to sit for another test in which 70% of them failed again. Find the percentage of students who failed in both tests.

A. 10%

B. 12%

C. 18%

D. 28%

E. 30%

Label 28 If $\frac{x+3y}{2x-y} = \frac{2}{3}$, then $\frac{x-y}{x+y} =$

A. $\frac{5}{6}$

B. $\frac{3}{5}$

C. $\frac{3}{5}$

D. $\frac{3}{4}$

E. $\frac{5}{6}$

Label 29 Suppose y is partly constant and partly varies inversely as x . When $x = 1$, $y = 7$ and when $x = 3$, $y = 3$. Find y when $x = 2$.

A. 2.5

B. 3.5

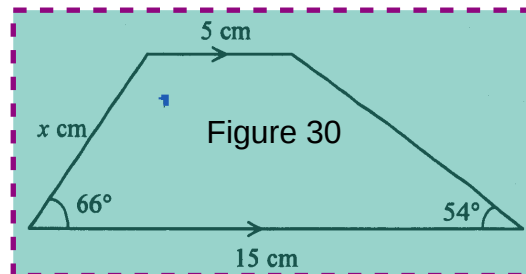
C. 4

D. 5

E. 6.5

Label 30 In the figure, find x correct to 3 significant figures.

- A. 8.86
- B. 9.34
- C. 9.48
- D. 10.7
- E. 11.3

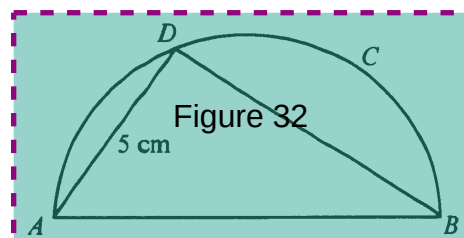


Label 31 Ship A is 8 km due north of a light house L and ship B is 6 km due east of L . Find the bearing of B from A .

- A. N53.1°W (correct to the nearest 0.1°)
- B. N36.9°W (correct to the nearest 0.1°)
- C. N36.9°E (correct to the nearest 0.1°)
- D. S53.1°E (correct to the nearest 0.1°)
- E. S36.9°E (correct to the nearest 0.1°)

Label 32 In the figure, $ABCD$ is a semicircle, $AB:BD = 4:3$. Find AB correct to the nearest 0.1 cm.

- A. 3.7 cm
- B. 7.6 cm
- C. 10.7 cm
- D. 13.0 cm
- E. 14.3 cm

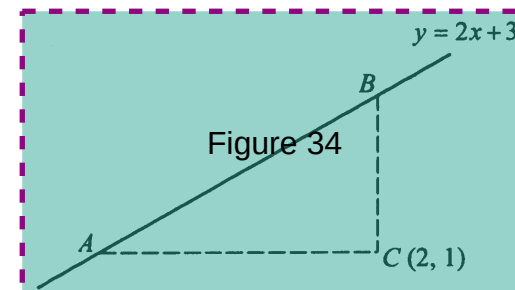


Label 33 If the straight lines $x - 2y + 5 = 0$ and $ax - y + 1 = 0$ intersect at $(1, b)$, find a and b .

- A. $a = -4, b = -3$
- B. $a = -1, b = 0$
- C. $a = 1, b = 3$
- D. $a = 2, b = -3$
- E. $a = 2, b = 3$

Label 34 In the figure, A, B and C are points on a rectangular coordinate plane. AC and BC are parallel to the x -axis and y -axis respectively. If the coordinates of C are $(2, 1)$ and the equation of the straight line AB is $y = 2x + 3$, find the distance between A and B .

- A. $\sqrt{5}$
- B. $\frac{3\sqrt{5}}{2}$
- C. $\sqrt{37}$
- D. $3\sqrt{5}$
- E. $\sqrt{65}$



Label 35 Two cards are drawn randomly from five cards numbered 1, 2, 3, 4 and 5 respectively. Find the probability that the sum of the two numbers drawn is even.

- A. $\frac{1}{2}$
- B. $\frac{2}{5}$
- C. $\frac{3}{10}$
- D. $\frac{7}{10}$
- E. $\frac{13}{25}$

Label 36 A bag contains 2 black balls and 3 white balls. A boy randomly draws balls from the bag one at a time (without replacement) until a white ball appears. Find the probability that he will make at least 2 draws.

- A. $\frac{2}{5}$
- B. $\frac{3}{5}$
- C. $\frac{1}{10}$
- D. $\frac{3}{10}$
- E. $\frac{7}{10}$

Section B

Label 37 If $\log x^2 = (\log x)^2$, then $x =$

- A. 1
- B. 10
- C. 100
- D. 1 or 10
- E. 1 or 100

Label 38 If $a > b$, which of the following must be true?

- I. $-a < -b$
- II. $a + b > b$
- III. $a^2 > b^2$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I, II and III

Label 39 If a , b are distinct real numbers and $\begin{cases} a^2 + 4a + 1 = 0 \\ b^2 + 4b + 1 = 0 \end{cases}$, find $a^2 + b^2$

- A. 1
- B. 4
- C. 14
- D. 16
- E. 18

Label 40 Suppose the graph of $y = x^2 - 2x - 3$ is given. In order to solve the quadratic equation $2x^2 - 6x - 3 = 0$, which of the following straight lines should be added to the given graph?

- A. $y = 4x$
- B. $y = x - \frac{3}{2}$
- C. $y = -x + \frac{3}{2}$
- D. $y = 2x - 3$
- E. $y = -2x + 3$

Label 41 Find the mean deviation of the five numbers 0, 3, 4, 6 and 7

- A. 0
- B. $\frac{3}{2}$
- C. $\frac{\sqrt{10}}{2}$
- D. 2
- E. $\sqrt{6}$

Label 42 For $0^\circ \leq x \leq 360^\circ$, how many roots does the equation $\cos^3 x = \cos x$ have?

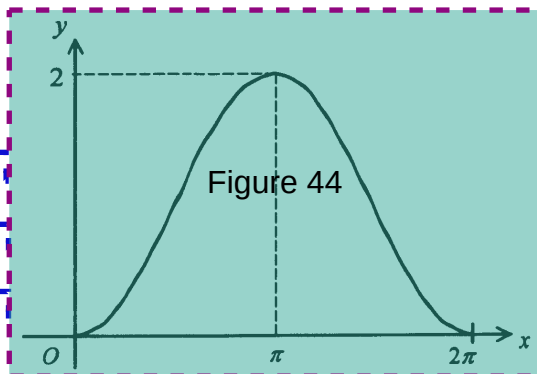
- A. 2
- B. 3
- C. 4
- D. 5
- E. 6

Label 43 If $\tan(90^\circ - \theta) = 2$, then $\frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta} =$

- A. 2
- B. $\frac{1}{2}$
- C. $\frac{1}{\sqrt{5}}$
- D. $-\frac{1}{2}$
- E. -2

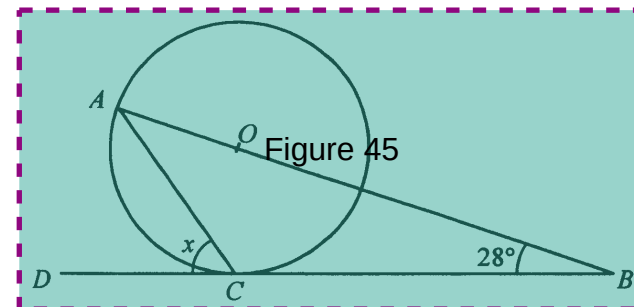
Label 44 The figure shows the graph of the function

- A. $y = \sin \frac{x}{2}$
- B. $y = 2 \sin x$
- C. $y = 1 + \sin x$
- D. $y = 1 + \cos x$
- E. $y = 1 - \cos x$



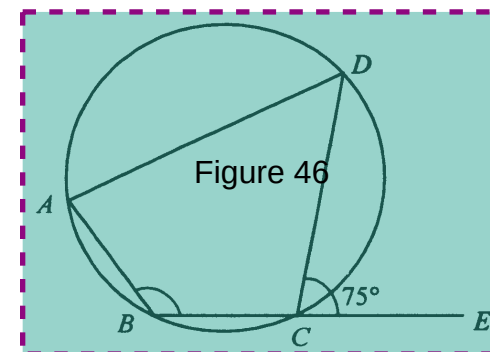
Label 45 In the figure, O is the centre of the circle, AOB is a straight line and BCD is the tangent to the circle at C . Find x .

- A. 50°
- B. 53°
- C. 36°
- D. 39°
- E. 62°



Label 46 In the figure, $\widehat{AB} = \widehat{BC} = \frac{1}{2} \widehat{CD}$. Find $\angle ABC$.

- A. 100°
- B. 105°
- C. 112.5°
- D. 130°
- E. 150°



Label 47 $\frac{1-x}{x^2+4x-5} + \frac{x-1}{x+1} =$

A. $\frac{x^2+3x-6}{(x+1)(x+5)}$

B. $\frac{x^2+5x-4}{(x+1)(x+5)}$

C. $\frac{(x+4)(x-1)}{(x+1)(x+5)}$

D. $\frac{(x-1)(x-4)}{(x+1)(x-5)}$

E. $\frac{(x-1)(x-6)}{(x+1)(x-5)}$

Label 48 Let $f(x) = x^3 + 2x^2 + ax + b$. If $f(x)$ is divisible by $x+1$ and $x-2$, $f(x)$ can be factorized as

A. $(x-1)(x+1)(x-2)$

B. $(x+1)^2(x-2)$

C. $(x-3)(x+1)(x-2)$

D. $(x+3)(x+1)(x-2)$

E. $x(x+1)(x-2)$

Label 49 The shaded region in the figure represents the solution of one of the following systems of inequalities. Which is it?

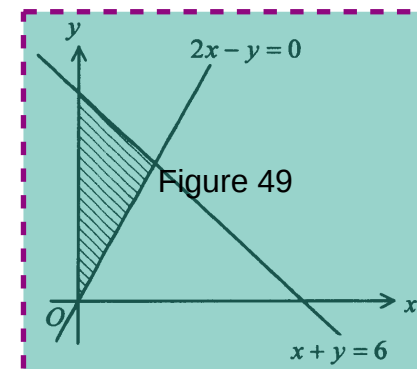
A. $\begin{cases} 2x-y \leq 0 \\ x+y \leq 6 \\ x \geq 0 \end{cases}$

B. $\begin{cases} 2x-y \leq 0 \\ x+y \leq 6 \\ y \geq 0 \end{cases}$

C. $\begin{cases} 2x-y \leq 0 \\ x+y \geq 6 \\ y \geq 0 \end{cases}$

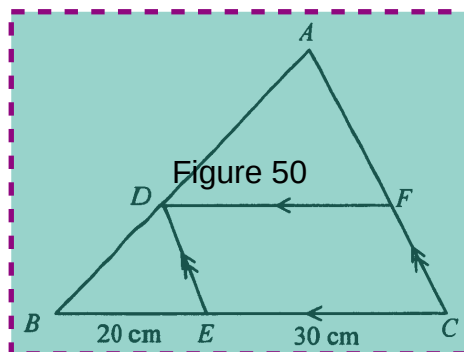
D. $\begin{cases} 2x-y \geq 0 \\ x+y \leq 6 \\ y \geq 0 \end{cases}$

E. $\begin{cases} 2x-y \geq 0 \\ x+y \geq 6 \\ x \geq 0 \end{cases}$



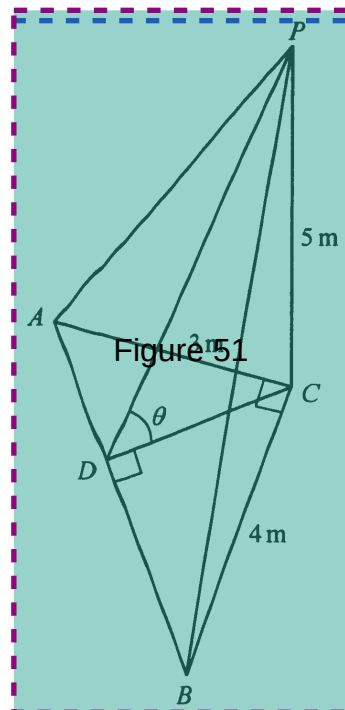
- Label 50 In the figure, ADB , BEC and CFA are straight lines. If the area of $\triangle ABC$ is 225 cm^2 , find the area of the parallelogram $DECF$.

- A. 81 cm^2
 B. 108 cm^2
 C. 126 cm^2
 D. 135 cm^2
 E. 162 cm^2



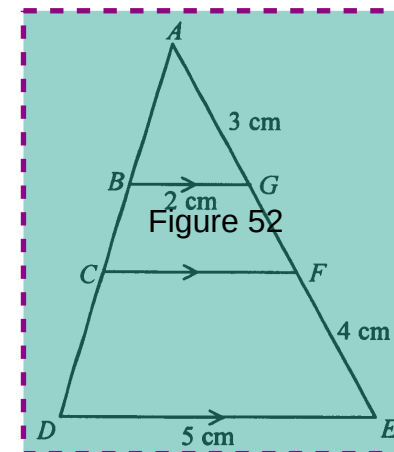
- Label 51 In the figure, PC is a vertical pole standing on the horizontal ground ABC . D is a point on line AB . If $\angle BCA = \angle CDB = 90^\circ$, $AC = 3 \text{ m}$, $BC = 4 \text{ m}$ and $PC = 5 \text{ m}$, find $\tan \theta$.

- A. $\frac{12}{25}$
 B. $\frac{16}{25}$
 C. $\frac{25}{16}$
 D. $\frac{25}{12}$
 E. $\frac{25}{9}$



- Label 52 In the figure, $ABCD$ and $AGFE$ are straight lines. Find CF .

- A. 4 cm
 B. 3 cm
 C. $\frac{7}{2} \text{ cm}$
 D. $\frac{5}{2} \text{ cm}$
 E. $\frac{7}{3} \text{ cm}$



- Label 53 Consider the circle $x^2 + y^2 - 8x - 6y + 21 = 0$. Find the equation of the chord whose mid-point is $(5, 2)$.

- A. $9x + 5y - 55 = 0$
 B. $3x + 4y - 23 = 0$
 C. $x + y - 7 = 0$
 D. $x - y + 3 = 0$
 E. $x - y - 3 = 0$

Label 54 In the figure, the inscribed circle of $\triangle OPQ$ touches PQ at R . Find the coordinates of R .

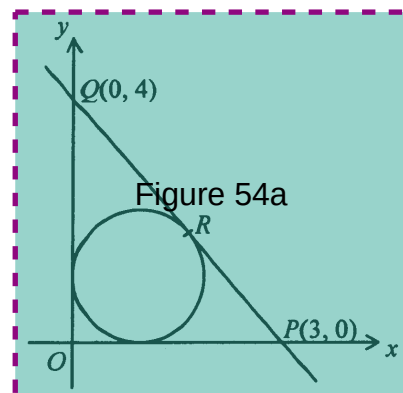
A. $\left(\frac{3}{2}, 2\right)$

B. $\left(\frac{6}{5}, \frac{12}{5}\right)$

C. $\left(\frac{9}{5}, \frac{8}{5}\right)$

D. $\left(\frac{9}{7}, \frac{16}{7}\right)$

Figure 54b $\left(\frac{12}{7}, \frac{12}{7}\right)$



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